bimm143_lab10

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Table of contents

1. Introduction to the RCSB Protein Data Bank (PDB) 1
2. Visualizing the HIV-1 protease structure 4
Bio3D Package for Structural Bioinformatics 6

Predicting functional motions of a single structure **8**

1. Introduction to the RCSB Protein Data Bank (PDB)

```r
library(dplyr)
```

```
Warning: package 'dplyr' was built under R version 4.5.2


Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.5.2
```

```
Warning: package 'tibble' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

Warning: package 'purrr' was built under R version 4.5.2

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v forcats   1.0.1     v readr     2.1.6
v ggplot2   4.0.2     v stringr   1.6.0
v lubridate 1.9.4     v tibble    3.3.1
v purrr     1.2.1     v tidyr     1.3.2

-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
pbd_data <- read.csv("pdb_stats.csv")
head(pbd_data)
```

```
           Molecular.Type   X.ray     EM    NMR Integrative Multiple.methods
1          Protein (only) 178,795 21,825 12,773         343              226
2 Protein/Oligosaccharide  10,363  3,564     34           8               11
3              Protein/NA   9,106  6,335    287          24                7
4     Nucleic acid (only)   3,132    221  1,566           3               15
5                   Other     175     25     33           4                0
6  Oligosaccharide (only)      11      0      6           0                1
  Neutron Other   Total
1      84    32 214,078
2       1     0  13,981
3       0     0  15,759
4       3     1   4,941
5       0     0     237
6       0     4      22
```

The commas in these numbers leads to the numbers here being read as characters.

```r
library(readr)
pbd_data_2<-read_csv("pdb_stats.csv")
```

```
Rows: 6 Columns: 9
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): Molecular Type
dbl (4): Integrative, Multiple methods, Neutron, Other
num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(pbd_data_2)
```

```
# A tibble: 6 x 9
  `Molecular Type`    `X-ray`    EM   NMR Integrative `Multiple methods` Neutron
  <chr>                 <dbl> <dbl> <dbl>       <dbl>               <dbl>   <dbl>
1 Protein (only)       178795 21825 12773         343                 226      84
2 Protein/Oligosacch~   10363  3564    34           8                  11       1
3 Protein/NA             9106  6335   287          24                   7       0
4 Nucleic acid (only)    3132   221  1566           3                  15       3
5 Other                   175    25    33           4                   0       0
6 Oligosaccharide (o~      11     0     6           0                   1       0
# i 2 more variables: Other <dbl>, Total <dbl>
```

```r
xrayem<-sum(c(pbd_data_2$`X-ray`, pbd_data_2$`EM`))
xray<-sum(pbd_data_2$`X-ray`)
em<-sum(pbd_data_2$EM)
total<-sum(pbd_data_2$`Total`)
percentxray <- xray/total
percentem<-em/total
percentxrayem<-100*(xrayem/total)
percentxrayem
```

```
[1] 93.7892
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
> ans: 93.8 for X-ray and EM total.
```

```r
100*(proteinstructures<-pbd_data_2[1,9]/total)
```

```
     Total
1 85.96889
```

Q2: What proportion of structures in the PDB are protein?

```
> ans: 85.96%
```

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

```
> ans: 1,173
```

2. Visualizing the HIV-1 protease structure

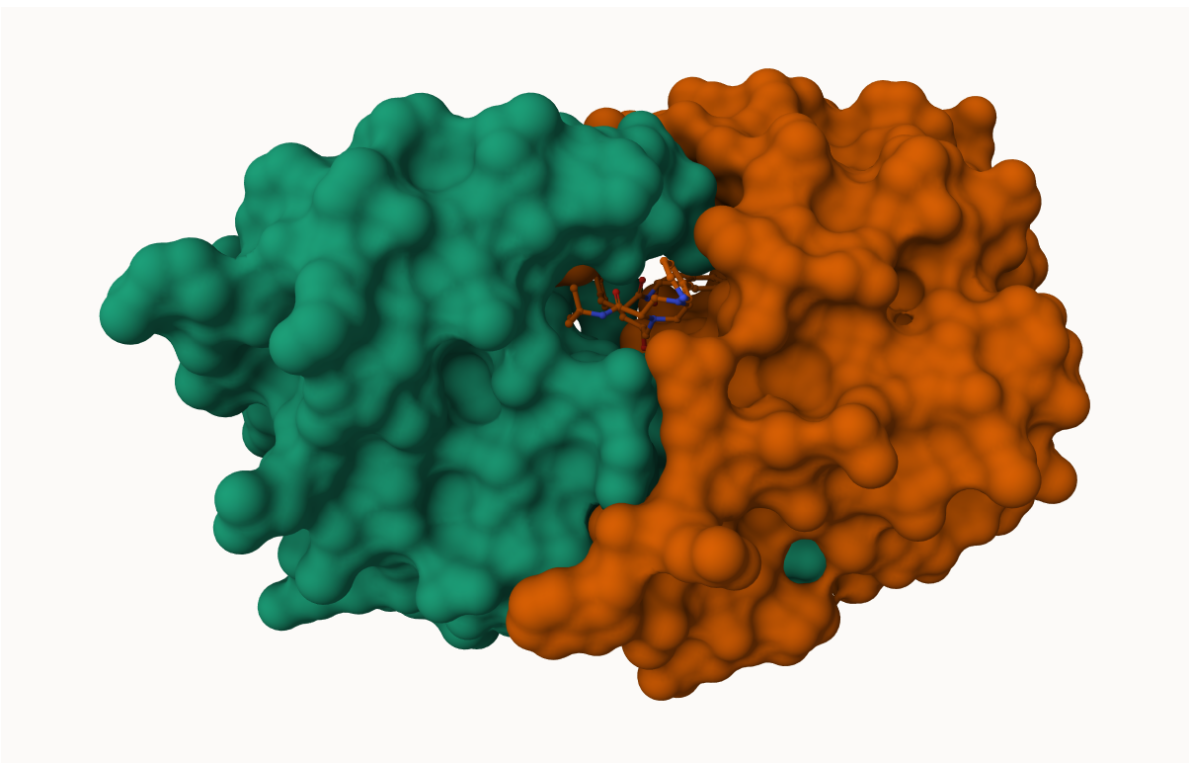


Figure 1: My first image of HIV-Pr with surface display showing ligand binding

Figure 2: My second image of HIV-Pr with surface display showing ligand binding

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

```
> ans: We see one atom because the other components of the molecule are more important and
```

Q5: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

```
> ans: 1608
```

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend "Ball & Stick" for these side-chains). Add this figure to your Quarto document.

```
> ans: in the quarto
```

Bio3D Package for Structural Bioinformatics

```r
library(bio3d)

pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```r
pdb
```

```
 Call:  read.pdb(file = "1hsg")

   Total Models#: 1
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 172  (residues: 128)
     Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
##MK1 is the ligand
```

```r
head(pdb$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
```

```
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

```r
library(bio3dview)
library(NGLVieweR)

#view.pdb(pdb)

#view.pdb(pdb) |>
  #setSpin()

#sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation

  #view.pdb(pdb, cols=c("navy","teal"),
         #highlight = sele,
         #highlight.style = "spacefill") |>
  #setRock()
```

Q7: How many amino acid residues are there in this pdb object?

> ans: 198

Q8: Name one of the two non-protein residues?

> ans: HOH

Q9: How many protein chains are in this structure?

> ans: 2

Predicting functional motions of a single structure

```r
adk <- read.pdb("6s36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

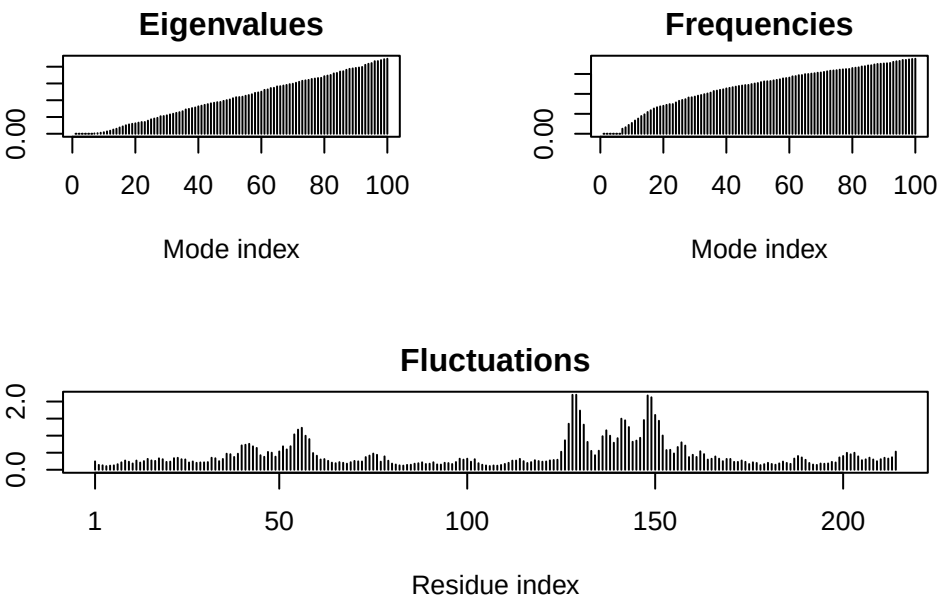
```r
adk
```

```
 Call:  read.pdb(file = "6s36")

   Total Models#: 1
     Total Atoms#: 1898,  XYZs#: 5694  Chains#: 1  (values: A)

     Protein Atoms#: 1654  (residues/Calpha atoms#: 214)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 244  (residues: 244)
     Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]

   Protein sequence:
      MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
      DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
      VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
# Perform flexiblity prediction
m <- nma(adk)
```

```
 Building Hessian...        Done in 0.036 seconds.
 Diagonalizing Hessian...   Done in 0.77 seconds.
```

```r
plot(m)
```




```
#mktrj(m, file="adk_m7.pdb")
```

```
#view.nma(m, pdb=adk)
```